## 4.2.2 Man-made fibres

Man-made fibres are fibres made by man with combination of natural products along with chemicals.

### A. Nylon

Nylon was the first synthetic fibre. In 1928 the Dupont Company decided to establish a fundamental research program. It was noticed that when a glass rod was taken out of one of the polyester stills the solution adhering to it stretched out into a solid filament. The filament could be stretched ever further and it did not go back to its original length. This stimulated the group to concentrate on textile fibres. The term nylon was chosen for the fibre and it was called the **Miracle fibre** for several years. The first nylon was referred to as type 6, 6.



Nylon

### i. Manufacturing process:

Nylon 6, 6 is a linear condensation polymer made from hexa methylene diamine and adipic acid. Specific amounts of the two chemicals are combined in solution to form nylon salt. The salt is purified, polymerized, extruded in ribbon form, and chipped into small flakes or pellets.

The polymer chips are melted by heat in an autoclave and pumped to the spinneret. The hot syrupy solution is pumped through the spinneret. It emerges in strands which can be stretched like warm taffy. The size of the fibre is determined by the size of the holes and the speed with which the fibre is withdrawn from the spinneret. The fibres are cooled by air blown across them. By drawing process either filament or staple fibres are prepared.

### ii. Properties:

| | | |
|---|---|---|
| Shape | : | Shape is controlled by the manufacturer, filaments are uniform and long. |
| Luster | : | Bright to Dull |
| Elastic Recovery | : | 100% |
| Elongation | : | Good |
| Resiliency | : | Good |
| Density | : | Good |
| Moisture absorption | : | 8 % |
| Dimensional stability | : | Excellent |
| Acids | : | Resistance is poor |
| Alkalis | : | Good resistance |
| Sunlight | : | Generally affects |
| Insects | : | Normally damages |
| To flame | : | Flammable |

### iii. Uses:

- Nylon is used in women's stocking or hosiery.
- It is also used as material for producing socks, swimwear, shorts, track pants, active wear, draperies and bedspreads.



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